

PGx Risk Dashboard — Hybrid Serverless Architecture

Tier 1: Frontend (S3)

S3 Static Frontend

Tab 1: Patient Input
Tab 2: Risk Score
Tab 3: What-If
Tab 4: Visualizations
Tab 5: PGx Card

Tier 2: AWS

API Gateway
(HTTPS/REST)

CORS: enabled
No auth (demo)

JSON

Tier 3: Serverless Compute

AWS Lambda
(Docker / ECR)

/risk
/causal-importance
/visualizations
/pgx-card

models

CPIC

schemas

Tier 4: Container (619 MB total, ECR)

Per-Density-Bin Models (up to 84)
CatBoost + XGBoost + XGBoost-RF
3 algorithms × 4 bins × 7 age bands (~469 MB)

CPIC DB Snapshot (March 2026)
573 gene-drug pairs · Level A/B only
12 PGx genes (~48 MB)

Feature Schemas + Training Medians
Per-bin thresholds · Imputation of Normality
(~18 MB)

Total: 619 MB (< 10 GB Lambda limit)

★ No patient data written to persistent storage — all computation ephemeral (HIPAA privacy-first; no BAA required)